HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2013)

Student No.: 14 Name: _____ Marks: / 24

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. Susan sells two cars for $80 080 each. She gains 30% on one and loses 30% on the other. After the transactions, Susan

   A. loses $15 840.

   B. gains $5 544.

   C. gains $10 296.

   D. has no gain and no loss.

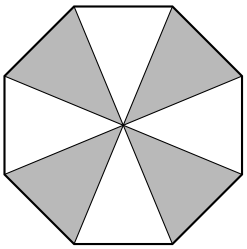
(DSE-2013 PII Q10)

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2. In the figure, the regular octagon is divided into eight identical isosceles triangles and four of them are shaded. The number of axes of reflectional symmetry of the octagon is



A. 2

B. 4

C. 8

D. 16

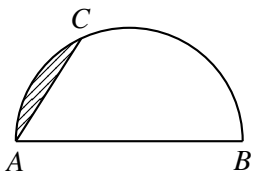
(DSE-2013 PII Q15)

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3. In the figure, the diameter of the semicircle ABC is 3 cm. If chord $AC = 2$ cm, find the area of the shaded region corrected to the nearest $0.01\,cm^2$.



A. $0.23\,cm^2$

B. $0.52\,cm^2$

C. $0.64\,cm^2$

D. $1.07\,cm^2$

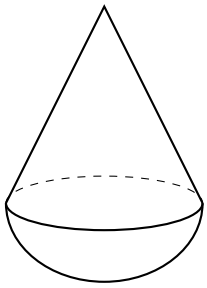
(DSE-2013 PII Q16)

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4. In the figure, the solid consists of a right circular cone and a hemisphere with a common base. The base radius and the height of the circular cone are 3 cm and 4 cm respectively. Find the total surface area of the solid.



A. $30\pi\,\text{cm}^2$

B. $33\pi\,\text{cm}^2$

C. $48\pi\,\text{cm}^2$

D. $51\pi\,\text{cm}^2$

(DSE-2013 PII Q17)

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5. If $0^\circ < x < 90^\circ$, which must be true?

I. $\tan x \tan(90^\circ - x) = 1$

II. $\sin x - \sin(90^\circ - x) < 0$

III. $\cos x + \cos(90^\circ - x) > 0$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2013 PII Q23)

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6. The coordinates of the points A and B are $(2,5)$ and $(4,-1)$ respectively. Let P be a moving point in the rectangular coordinate plane such that $AP = BP$. Find the equation of the locus of P.

A. $x-3y+3=0$

B. $x-3y-7=0$

C. $x-3y+13=0$

D. $3x+y-11=0$

(DSE-2013 PII Q24)

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7. The equation of the circle C is $2x^2+2y^2-4x+8y-5=0$. The coordinates of the points P and Q are $(-1,2)$ and $(4,0)$ respectively. Which of the following is/are true?

I. The radius of C is 5.

II. The midpoint of PQ lies outside C.

III. If G is the centre of C, then $\angle PGQ$ is acute angle.

A. I only

B. II only

C. I and III only

D. II and III only

(DSE-2013 PII Q25)

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8. If the mean and mode of nine numbers $14, 6, 4, 5, 7, 5, x, y, z$ are 8 and 14 respectively, then the median of these nine number is

A. 5.

B. 6.

C. 7.

D. 8.

(DSE-2013 PII Q27)

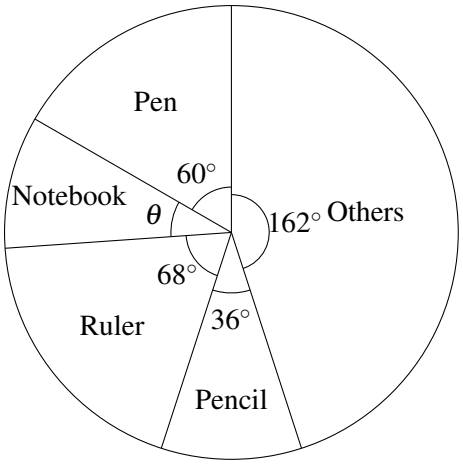
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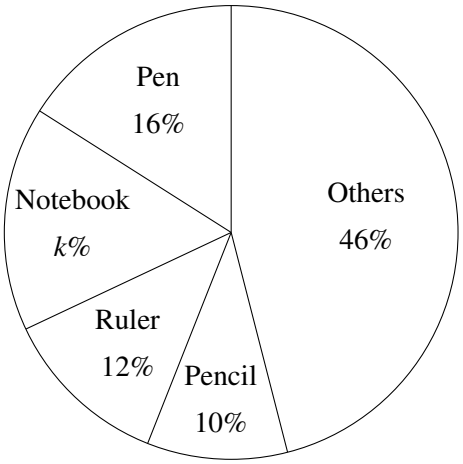
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9. The pie charts below show the distributions of the profits of stationery shop$X$ and stationery shop Y from the sales of stationery in a certain month. Which of the following must be true?

Distribution of the profits of stationery shop X



Distribution of the profits of stationery shop Y



A. In that month, the profit from the sales of pencils of stationery shop X is the same as that of stationery shop Y.

B. In that month, the total profit from the sales of pens and notebooks of stationery shop X is less than the total profit from the sales of rulers and pencils of the shop.

C. $k = 14$

D. $\theta = 36^\circ$

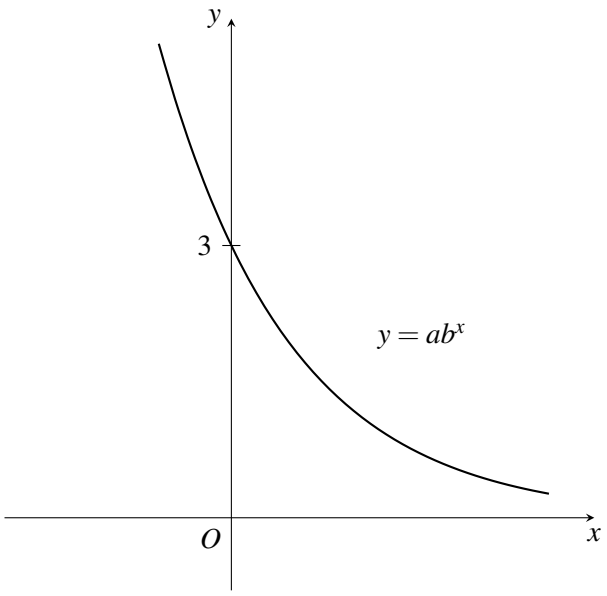
(DSE-2013 PII Q30)

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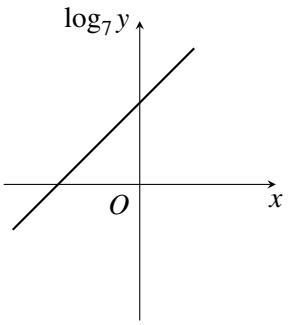
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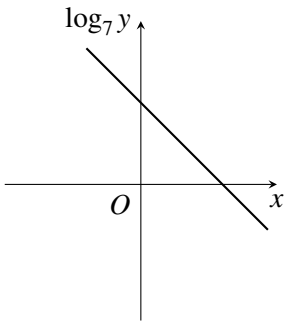
10. The figure below shows the graph of $y = ab^x$, where a and b are constants. Which of the following graphs may represent the relation between x and $\log_7 y$?



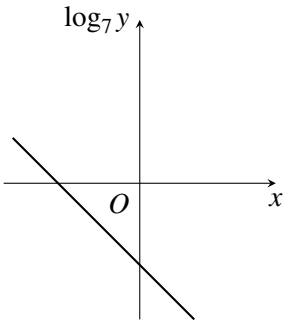
A.



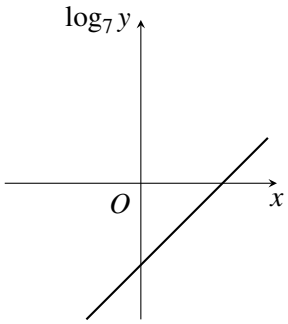
B.



C.



D.



(DSE-2013 PII Q32)

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11. $A00000E00001_{16} =$

A. $10 \times 16^{11} + 14 \times 16^5 + 17$

B. $11 \times 16^{11} + 15 \times 16^5 + 17$

C. $10 \times 16^{12} + 14 \times 16^6 + 272$

D. $11 \times 16^{12} + 15 \times 16^6 + 272$

(DSE-2013 PII Q33)

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12. If the height of a regular tetrahedron has height 2 cm. then the volume of the tetrahedron is

A. $2\,\text{cm}^3$

B. $\sqrt{3}\,\text{cm}^3$

C. $\sqrt{6}\,\text{cm}^3$

D. $3\sqrt{3}\,\text{cm}^3$

(DSE-2013 PII Q40)

I can finish _____ questions

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